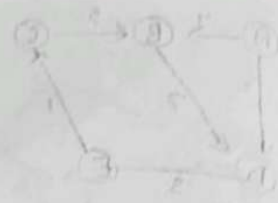


CO4-ASSESSMENT Tool Comparative Analysis



Linear vs Quadratic vs Double Hashing

Method	Primary clustering	Secondary clustering	cache performance
Linear Probing	High	No	Best
Quadratic Probing	No	yes	Good
Double hashing	No	No	Good

Best cache performance: Linear probing because it accesses consecutive memory locations, giving excellent spatial locality

Best overall collision resolution: Double hashing because it minimizes both primary and secondary clustering and generally gives fewer probes

Average Probes

Higher as load increases
Lower than linear
Lowest

Binary Heap vs Balanced BST

Operation	Binary Heap	Balanced BST
Insert	$O(\log n)$	$O(\log n)$
Extract-Min	$O(\log n)$	$O(\log n)$
Decrease-Key	$O(\log n)$	$O(\log n)$
Arbitrary Key update/search	$O(n)$	$O(\log n)$
Memory	$O(n)$, array-based	$O(n)$, extra pointers

which is more efficient?

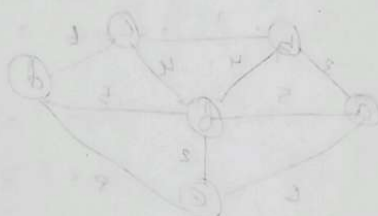
Binary heap can be stored in a simple array with no left/right-child pointers. A balanced BST requires extra memory for node pointers.

final choice for scheduled

If the Scheduler mainly extract-min + decrease key a balanced BST is better when arbitrary key update/search are frequent.

If memory efficiency and priority queue operations are the main concern, choose a binary heap.

Final
2/18/20



Complexity = $O(\log V)$
 $(E) \log(V) =$

Tree Vertex	Remaining Vertex	Illustration
$(-)$	(f, a) $(c, -)$ $(d, -)$ (e, a) $(b, -)$	
(a, a)	(c, a) $(d, -)$ (e, a) (b, a)	
(a, b)	(c, a) $(d, -)$ (e, a) (b, a)	
(a, c)	(d, a) (e, a) (b, a)	
(a, d)	(e, a) (b, a)	
(a, e)	(b, a)	
(a, b)		